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UNIT-10

Software Management

Software Project Management:

Software project management is an art and science of planning and leading software projects. It is a sub-discipline of project management in which software projects are planned, implemented, monitored and controlled. It is an essential part of software engineering. Good project management cannot guarantee project success however bad management usually results in project failure.

Software Project Manager:

A software project manager is a person who undertakes the responsibility of planning, executing and controlling the software project. Project managers are responsible for planning and scheduling project development. Software project management is differing from other type of project management due to following reasons:

i) The product is intangible in nature and it can be realized only, it cannot be seen or touched.
ii) There are no standard software processes.
iii) Large software projects are often One-off projects.

Project Management Activities:

1. Project Planning: Project planning is an organized and integrated management process, which focuses on activities required for successful completion of the project. Project planning involves breaking down the work into parts, assign those to project team members, allocate resources, and prepare solutions to problems.

Project planning sections:

i) **Introduction:** It describes objectives of the projects and sets out the constraints like time, budget etc.

ii) **Project organization:** It describes the people involved and their role in team.

iii) **Risk analysis:** It describes possible project risks.

iv) **Hardware and software resource requirement:** It specifies the hardware and software required to carry out the development.

v) **Work breakdown:** The project is breakdown into activities and identifies milestones and deliverables associated with each activity.

vi) **Project schedule:** It shows the dependencies between activities.

vii) **Monitoring and reporting mechanism:** This defines the management report that should be produced.

Types of project plan:

i) **Quality plan:** It describes the quality procedures and standards that will be used in a project.

ii) **Validation plan:** It describes the approach, resources and schedule used for system validation.

iii) **Configuration management plan:** It describes the configuration management procedures and structures to be used.

iv) **Maintenance plan:** It predicts the maintenance requirements of the system, maintenance costs and effort required.

v) **Staff development plan:** It describes how the skill and experience of the project team members will be developed.

2. Project Scheduling:

Project scheduling is the process of separating the total work involved in a project into separate activities and judging the time required to complete these activities. It is a detailed plan showing dates when each activity should start and finish, also it includes when and how much resource will be required.

Creating a project schedule includes four stages:

→ Constructing an ideal activity plan.
→ Risk analysis
→ Resource allocation
→ Schedule production.

Project schedules are usually represented as a set of charts showing the work breakdown, activities dependencies and staff allocations. Bar charts (for example Gantt chart) and activity networks are graphical notations that are used to illustrate the project schedule. Bar charts show who is responsible for each activity and when the activity is scheduled to begin and end. Activity networks show the dependencies between the different activities.

3. Risk Management:

Risk management is process of identifying risks that might affect the project schedule or quality of project being developed and taking action to avoid these risks. Risk management involves identifying and assessing major project risks to establish the probability that they will occur and the consequences for the project if that risk does arise.

There are mainly three categories of risks which are likely to occur in software projects:

4. Project Risk: Project risks influences the schedule of software project and that if occur, delay the development process. Example: staff turnover, hardware unavailability etc.

ii) **Technical risks:** Technical risks involves implementation, testing, and maintenance issue. Most technical risks appear due to the development team's insufficient knowledge about the project.

iii) **Business risk:** It affects the organization business that is developing the software. Business risks could be quite dangerous for the long-term sustainability of the business. Example: Technology change, product completion etc.

Risk Management Process:

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│ Risk Identification │ ──> │ Risk Analysis │ ──> │ Risk Planning │ ──> │ Risk Monitoring │ ──┐
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┌────────────────────┐ ┌──────────────┐ ┌──────────────┐ ┌────────────────┐ │
│ List of Potential │ │ Prioritized │ │ Risk Avoidance│ │ Risk │ │
│ Risks │ │ Risk List │ │and Contingency│ │ Judgement │ │
│ │ │ │ │ plans │ │ │ │
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Fig: Risk management process.

The risk management process is an iterative process which continues throughout the project. Once an initial set of plans are drawn up, the situation is monitored.

i) **Risk identification:** It is the first stage of risk management. It is concerned with discovering possible risk to the project. It also involves preparation of risk list.

ii) **Risk Analysis:** In this phase all identified risks are considered and made a judgement about problems causing risk in the project, identify probability of occurrence of risk and the impact of the risk. The probability of risks may be judged as very low (< 10%), low (10-25%), moderate (26-50%), high (51-75%), and very high (> 75%). Risks are tabulated in order according to the seriousness of the risk.

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iii) **Risk planning:** It is the process of consideration of the key risks that have been identified and formulation of strategies to manage the risk. The strategies to manage risk are:

- Avoidance Strategies (that tells reduction of arising risks).
- Minimization Strategies (that tells impact of risk will be reduced).
- Contingency Plans (Prepared for worst and have strategy to deal with it).

iv) **Risk monitoring:** Risk monitoring means regularly judging each of the identified risk to decide whether or not the risk is becoming more or less probable and whether the effect of risk have changed.

4) People Management:

The process of planning, organizing, directing, and controlling human resources is called people management. It is also called human resource management. People Management ensures that organizational, individual, and societal needs are satisfied. Human resource management includes all activities used to attract and retain employees.

5) Proposal Writing:

Project manager have to write the project proposal to win a contract from the customer. Proposal writing is a skill that acquire through practice and experience. Proposal should include the objective of the project and how it will be carried out. It also includes cost and schedule estimates.

⊛ Milestones and Deliverable:

Project milestones are the predictable outcome of an activity where some formal report of progress are addressed. Milestones are recognizable end points of software process activity, they may be short report of what has been completed and presented to the management. Milestones should represent the end of a distinct, logical stage in the project.